25901182 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBCA201 - INTRODUCTION TO MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

1. Distinguish between the four outcomes represented in a confusion matrix. [U] / [CO1]
2. Define centroid in K-Means clustering. [K] / [CO2]
3. Show why ML is important today. [U] / [CO1]
4. Explain the difference between hard margin and soft margin SVM. [U] / [CO2]
5. Describe logistic regression. [U] / [CO2]
6. Compare clustering and classification. [An] / [CO1]
7. Identify whether a given dataset is linearly separable or not. [U] / [CO2]

[3x5 = 15]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

8. Analyze how multiple factors improve prediction accuracy in multiple linear regression. [An] / [CO1]
9. Explain the challenges faced while implementing ML models in real-world scenarios. [An] / [CO2]
10. Build a Perceptron model for AND gate and draw the decision boundary. [A] / [CO2]
11. Explain the role of Machine Learning in image and speech recognition. [U] / [CO2]
12. Write the Naive Bayes classifier algorithm and its working. [A] / [CO2]
13. Identify the learning task: Disease positive or negative prediction. Justify. [U] / [CO1]

[5x4 = 20]